Asness, Frazzini and Pedersen (2019) move the lens from analysing if firm characteristics predict returns to whether market prices adequately reflect those characteristics. The shift matters as the allocative consequences of market efficiency operate through prices and a firm's cost of capital. While Asness at al. (2019) does find that investors do pay more for firms with higher quality characteristics, its explanatory power is weak.

The characteristics within the stock are responsible for its higher price. These include profitability, growth and safety, $Quality_{i,t} = z\,(Profitability_{i,t} + Growth_{i,t} + Safety_{i,t})$. Profitability includes measures like gross profits over assets, ROE, ROA, cash flow over assets, margins and low accruals. Growth measures the five-year change in residual profitability measures, while safety incorporates low beta, low leverage, low bankruptcy and low earnings volatility.

Asness et al. (2019) develop the factor portfolio 'Quality-minus-Junk' (QMJ) that uses a **Dynamic asset pricing model** with *time-varying* **growth**, **profitability** and **risk** variables (price-to-book ratios increasing linearly). Following Fama and French (1993), the QMJ portfolio goes long in **high-quality** stocks and shorts **low-quality** 'junk' ones, specifically, longing the top 30% and shorting the bottom 30%, with QMJ being formed separately within small and large firms. $QMJ = \frac{1}{2}(Small\ Quality - Small) + \frac{1}{2}(Big\ Quality - Big\ Junk)$ The QMJ factor averages the spread among small and large firms. The price of quality (the marginal amount investors pay for higher quality characteristics) varies over time (varying the price premium) and predicts future returns of the QMJ portfolio. The three quality characteristics exhibit persistence across 5 to 10 years, with profitability being the highest.

The **Gordons growth model** $\left(\frac{P}{B} = \frac{Profitability\ x\ (Payout-ratio)}{(required-return-growth)}\right)$ helps to explain QMJ construction. The QMJ factor portfolio allows for the *P/B* ratio to vary with the 'quality' characteristics across stocks and time, capturing how prices increase with quality through time. Additionally, prices are scaled by book equity to have the analysis explain the valuation multiples rather than absolute firm values. Consequently, this allows firms of different sizes to become comparable and increases stationarity dependent variables across the cross-section and over time.

C**ross-sectional regressions** of *P/B* on each stock's **quality score** finds that higher 'quality' is significantly associated with higher prices yet exhibits low **explanatory power** (0.10). However, controlling for size, past 12 months, industry and country, rises to (0.43). Tests to explain why follow, splicing the portfolio into deciles allows the consideration of the **value weighted return** of each portfolio. Additionally, several empirical tests ensue to investigate the low R^2.

Three tests are gradually ruled out, sequentially testing the following assumptions (1) *market prices use different quality characteristics than the study* (implies that the stocks classified as quality does not necessarily predict that the stocks have high returns, is ruled out). (2) *quality characteristic is correlated with risk factors not captured in the studies risk adjustments* (predicting that high-quality stocks should have low returns during distress periods, is ruled out, as <u>QMJ has high returns in high volatility periods</u>*)* and (3) *market prices fail to fully reflect the quality characteristics for reasons linked to behavioural finance or constraints (*high-quality stocks do not have high risk-adjusted returns*)*. Critically, QMJ returns are oddly high during market downturns.

The QMJ factor portfolio challenges risk-return explanations based on a covariance with market crises. QMJ exhibits a mild positive convexity, meaning that the portfolio benefits <u>from a mass 'flight to quality' during a crises.</u> Testing this behavioural explanation, Asness et al. (2019) use target prices (the expected stock price one year into the future) from the forecasts created by **equity analyst** and find that the analysts estimated **implied return expectation**, (the target price divided by current actual price) on average, <u>is lower for high-quality stocks than junk</u>. Additionally, the equity analysts are too optimistic about junk stocks, forecasting earnings above realised earnings.

The current price of quality (defined as the **cross-sectional regression coefficients)** reflects how the price of quality varies across time. Hence, the *time-series of cross-sectional regression coefficients reflects how the pricing of quality changes over time*. <u>The price of quality was low during periods</u>; Feb2000, 1987crash, 2007-09crash; following these events <u>the price of quality increased during periods</u>; late1990gulfwar, late2002enron/WorldCom, early2009heightbankingcrises. As such, the extreme events display how prices and returns are connected. Specifically, the price of quality negatively predicts the future returns of the QMJ portfolio. The QMJ portfolio exhibits negative MKT, HML, SMB exposures with positive alphas and small residual risk.

The value factor portfolio 'High-minus-Low' (HML) and QMJ are negatively correlated, being opposite. The QMJ factor portfolio *buys/sells <u>based on quality</u> irrespective of price*, while HML buys *<u>based on</u> a stocks price, irrespective of quality*. The portfolios can be combined to obtain the **'quality at a reasonable price' (QARP) factor**. Theoretically, the QMJ portfolio aligns with findings that the information from financial statements improves value investing. Crucially, *quality stocks earn higher returns yet are safer not riskier.*

The mathematical model of QMJ illustrates how, "*a company is worth its current book equity, plus the present value of profits it can sustainably earn above the cost of that equity, with faster growth raising value, the temporary accounting profits receive little weight as greater risk lowers value* "Additionally, a company's sustainable profits, and sustainable growth raises its 'quality' characteristic, while greater risk lowers its value. Financial markets are assumed to be forward looking, with 'high quality' being persistent, coincides with both prices relation to a future quality characteristic and its predictable component. **Large stocks** are more liquid and have less liquidity risk than small firms, thus higher prices and lower required returns.